

① a) $5x - 7 \leq 0 \rightarrow x \leq \frac{7}{5}$ b) $7 - 3x < 10 \rightarrow -3x < 3 \rightarrow x > -1$ c) $51 \leq 6x + 15 \rightarrow 6x \geq 36 \rightarrow x \geq 6$

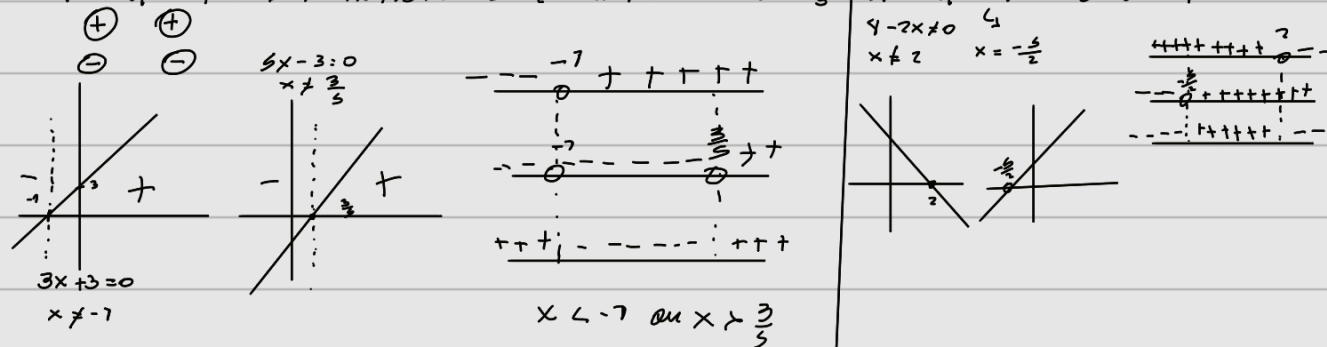
d) $76 - 4x \geq 77x - 29 \rightarrow 76x \leq 105 \rightarrow x \leq \frac{105}{76}$ e) $\frac{2x}{3} + 6 \geq \frac{x}{5} + 9 \rightarrow \frac{2x}{3} - \frac{x}{5} \geq 3 \rightarrow \frac{10x - 3x}{15} \geq 3 \rightarrow 7x \geq 45 \rightarrow x \geq \frac{45}{7}$

f) $3x - 21 > 9x - 21 \rightarrow x < 0$ g) $\frac{7x-4}{4} + \frac{7-3x}{2} > \frac{x-1}{3} - 1 \Rightarrow \frac{3(x-2) - 4(x-4)}{3 \cdot 4} \Rightarrow x < 10$

h) $\frac{x+70}{5} > -x+6 \Rightarrow \frac{x+70}{5} + x - 6 > 0 \Rightarrow \frac{x+70+5x-30}{5} > 0 \Rightarrow \frac{6x+40}{5} > 0 \Rightarrow x > -\frac{20}{3}$

i) $\frac{7-2x}{3} + \frac{x-2}{6} > \frac{x+3}{2} - 1 \Rightarrow \frac{-5x+3}{6} > \frac{x+1}{2} \Rightarrow \frac{-5x+3}{6} > \frac{x+3-3}{2} \Rightarrow -\frac{x}{2} > \frac{x+1}{2} \Rightarrow x < -\frac{1}{2}$

② a) $|3x+3|/|5x-3| > 0 \Rightarrow 15x^2 - 9x + 9 > 0 \Rightarrow S = \{x \in \mathbb{R} \mid x < -1 \text{ ou } x > \frac{3}{5}\}$ b) $|4-2x|/|5+2x| < 0 \Rightarrow S = \{x \in \mathbb{R} \mid x < -\frac{5}{2}\}$



c) $|3x+2|/|-2x+4| \cdot |x-6| < 0$ d) $|5-2x|/|-2x-2| \leq 0$

③ a) $\frac{2x+1}{x+2} > 0 \Rightarrow x \neq -2$ b) $\frac{3x-2}{3-2x} \leq 0 \Rightarrow 3x-2=0 \Rightarrow x = \frac{2}{3}$ c) $\frac{3-4x}{5x+7} \geq 0 \Rightarrow x = -\frac{7}{5}$

d) $\frac{-3-2x}{3x+1} \leq 0 \Rightarrow x = -\frac{3}{2}$ e) $\frac{3x-2}{7-x} \leq 3 \Rightarrow 3x-2-3 \leq 0 \Rightarrow 3x-5 \leq 0 \Rightarrow x \leq \frac{5}{3}$

④ a) $\frac{2x-3}{x-7} \leq 2 \Rightarrow \frac{2x-3-2x+14}{x-7} \geq 0 \Rightarrow \frac{11}{x-7} \geq 0 \Rightarrow x > 7$ b) $\frac{3x-2}{7-x} \leq 3 \Rightarrow \frac{3x-2-3 \cdot 7 + 21}{7-x} \leq 0 \Rightarrow \frac{6x-19}{7-x} \leq 0 \Rightarrow x \leq \frac{19}{6}$

c) $\frac{5x-3}{3x-4} > -7 \Rightarrow \frac{5x-3}{3x-4} + 7 > 0 \Rightarrow \frac{5x-3+3x-28}{3x-4} > 0 \Rightarrow \frac{8x-31}{3x-4} > 0 \Rightarrow x > \frac{31}{8}$

d) $\frac{x-7}{x+1} \geq 3 \Rightarrow \frac{x-7-3x-3}{x+1} \geq 0 \Rightarrow \frac{-2x-10}{x+1} \geq 0 \Rightarrow \frac{x+5}{x+1} \leq 0 \Rightarrow -5 < x < -1$

e) $\frac{6x}{x+3} \leq 5 \Rightarrow \frac{6x-5x-15}{x+3} \leq 0 \Rightarrow \frac{x-15}{x+3} \leq 0 \Rightarrow x \leq 15$

f) $\frac{x+7}{x-2} \geq 4 \Rightarrow \frac{x+7-4x+8}{x-2} \geq 0 \Rightarrow \frac{-3x+15}{x-2} \geq 0 \Rightarrow \frac{x-5}{x-2} \leq 0 \Rightarrow 2 < x \leq 5$

⑤ a) $-2 \leq \frac{x+3}{1} \leq 9 \Rightarrow \frac{x+3}{1} \leq 9 \Rightarrow x \leq 6$ b) $\frac{1}{6} \leq \frac{8}{3} - \frac{5x}{2} \leq \frac{7}{6} \Rightarrow \frac{1}{6} \leq \frac{16-15x}{6} \leq \frac{7}{6} \Rightarrow 1 \leq 16-15x \leq 7 \Rightarrow -15 \leq -15x \leq -9 \Rightarrow 1 \leq x \leq \frac{3}{5}$

c) $-\frac{3}{2} \leq \frac{2x-5}{3} < \frac{1}{6} \Rightarrow \frac{2x-5}{3} \geq -\frac{3}{2} \Rightarrow 2x-5 \geq -\frac{9}{2} \Rightarrow 2x \geq -\frac{1}{2} \Rightarrow x \geq -\frac{1}{4}$

d) $\frac{1}{2} \leq \frac{5x-2}{4} < \frac{7}{3} \Rightarrow \frac{5x-2}{4} \geq \frac{1}{2} \Rightarrow 5x-2 \geq 2 \Rightarrow 5x \geq 4 \Rightarrow x \geq \frac{4}{5}$

e) $\frac{5x-2}{4} < \frac{7}{3} \Rightarrow 5x-2 < \frac{28}{3} \Rightarrow 5x < \frac{34}{3} \Rightarrow x < \frac{34}{15}$

8) a) $\frac{1}{x-4} < \frac{2}{x+3} \rightarrow \frac{1}{x-4} - \frac{2}{x+3} < 0 \rightarrow \frac{x+3-2(x-4)}{(x-4)(x+3)} < 0 \rightarrow \frac{-x+11}{(x-4)(x+3)} < 0$

b) $\frac{x+1}{x+2} > \frac{x+3}{x+4} \rightarrow \frac{(x+1)(x+4) - (x+3)(x+2)}{(x+2)(x+4)} > 0 \rightarrow \frac{-2}{(x+2)(x+4)} > 0$

c) $\frac{1}{x-1} + \frac{2}{x-2} - \frac{3}{x-3} < 0 \rightarrow \frac{2 \cdot (2x-3)}{(x-1)(x-2)(x-3)} < 0$

d) $\frac{x+5}{3x+2} - \frac{x-2}{3x+5} \leq 0 \rightarrow \frac{(3x+5)(x+5) - (x-2)(3x+2)}{(3x+2)(3x+5)} \leq 0$

9) a) $x^2 - 2x + 2 > 0$ $\Delta < 0$, portanto para todo x haverá um valor y positivo.

b) $x^2 - 2x + 1 > 0$ $x=1$ $\Delta = 0$ $S = \mathbb{R} - \{1\}$

c) $x^2 - 2x > 0 \rightarrow x(x-2) > 0$ $S = x \in (-\infty, 0) \cup (2, +\infty)$

e) $-x^2 + x + 6 > 0$ $x_1 = 3$ $x_2 = -2$ $S = x \in (-\infty, -2) \cup (3, +\infty)$

10) a) $(1-4x^2) \cdot (2x^2+3x) > 0$

b) $x \cdot (2x+3) \geq 0$ $x_1 = 0$ $x_2 = -\frac{3}{2}$

d) $-2x^2 + 3x + 2 \geq 0$ $x_1 = 2$ $x_2 = -\frac{1}{2}$ $S = x \in [-\frac{1}{2}, 2]$

f) $8x^2 - 74x + 3 \leq 0$ $x_1 = \frac{3}{4}$ $x_2 = \frac{7}{4}$ $S = x \in [\frac{3}{4}, \frac{7}{4}]$

c) $(x^2 - x + 6)(-x^2 - 2x + 3) \geq 0$

d) $x^2 - x + 6 = 0$ $\Delta < 0$, portanto para todo x

e) $-x^2 - 2x + 3 = 0$ $x_1 = -3$ $x_2 = 1$ $S = x \in [-3, 1]$

b) $(2x^2 - 7x + 6)(2x^2 - 7x + 5) \leq 0$

c) $2x^2 - 7x + 6 = 0$ $x_1 = \frac{3}{2}$ $x_2 = 2$

d) $2x^2 - 7x + 5 = 0$ $x_1 = \frac{5}{2}$ $x_2 = 1$ $S = x \in [1, \frac{5}{2}] \cup [\frac{5}{2}, 2]$

11) a) $\frac{4x^2+x-5}{2x^2-3x-2} > 0$

b) $\frac{9x^2+9x-2}{3x^2+7x+2} \leq 0$

c) $\frac{2-3x}{2x^2+3x-2} < 0$

d) $4x^2+x-5 = 0$ $x_1 = \frac{5}{4}$ $x_2 = -1$

e) $9x^2+9x-2 = 0$ $x_1 = \frac{2}{3}$ $x_2 = -\frac{1}{3}$

f) $3x^2+7x+2 = 0$ $x_1 = -\frac{1}{3}$ $x_2 = -2$

g) $2x^2+3x-2 = 0$ $x_1 = \frac{1}{2}$ $x_2 = -2$

h) $2x^2+3x-2 = 0$ $x_1 = \frac{1}{2}$ $x_2 = -2$

12) 12 de largura e 25 de comprimento
300m² total.

$10 \cdot x < 300 \cdot \frac{2}{3} \rightarrow 10x < 200 \rightarrow x < 20$ O comprimento máximo é de 20 metros

13) $30 + \frac{2}{3}l + l \leq 158 \rightarrow \frac{90+2l+3l}{3} \leq 158 \rightarrow \frac{90+5l}{3} \leq 158 \rightarrow \frac{5l-384}{3} \leq 0 \rightarrow l \leq 76,8 \text{ km}$

14) $f(x) = 3,44 + 0,9x$ $20 = 3,44 + 0,9x \rightarrow x = 18,4$ $30 = 3,44 + 0,9x \rightarrow x = 29,5$ $18,4 \leq x \leq 29,5$